Dear Editor and Reviewers,

We truly appreciate all your valuable comments. After a careful review, we have adopted the observations in our revised manuscript, entitled "*Regression-Based Explainable Deep Learning for Estimating Hamiltonian Parameters from Magnetic Nanodot Images*". Our point-to-point responses to the reviewers' comments are given below:

REVIEWER 1

Point 1. *The authors demonstrate a high prediction accuracy for the DMI parameter using only the out-of-plane spin component (sz) as the input image. I think this may be possible because the direction of the DMI vector is fixed and only a limited range of the DMI parameter (for only positive values) is considered. Since the sz component does not contain the full information of the spin chirality, opposite signs of the DMI can potentially generate very similar out-of-plane magnetic textures with opposite in-plane winding directions. Therefore, it would be valuable if the authors could discuss how the prediction performance would be affected when both positive and negative DMI values are included simultaneously in the dataset. This discussion would help clarify the applicability and limitations of the proposed approach for more general DMI systems.*

Response. We thank the reviewer for this insightful observation, with which we fully agree. The reviewer is correct that the out-of-plane component s_z does not encode the in-plane spin chirality, and that opposite signs of the DMI generate nearly indistinguishable s_z textures with reversed in-plane winding. Restricting the network input to the s_z projection—and, consequently, sampling the DMI over non-negative values only—is a deliberate, physically motivated design choice, grounded in two complementary considerations:

1. *Experimental measurability.* The s_z map corresponds to the observable that is most directly accessible in practice. The most widely available magnetic imaging techniques—magnetic force microscopy (MFM) and the polar magneto-optical Kerr effect (MOKE)—are primarily sensitive to the out-of-plane stray field and magnetization component. Resolving the in-plane components (s_x, s_y) requires substantially more specialized modalities, such as spin-polarized scanning tunneling microscopy with in-plane sensitivity, Lorentz transmission electron microscopy, or electron holography. Training the framework on s_z there-

fore keeps the method consistent with the information realistically available from experiment.

2. *Architectural compatibility.* A single scalar field s_z defined over the two-dimensional lattice is naturally represented as a grayscale image, which maps directly onto the input format expected by the CNN and ViT backbones employed here. Providing the full spin-vector field would triple the number of input channels while adding precisely the in-plane information that is degenerate with respect to the DMI sign, as discussed below.

Regarding the reviewer's specific request—how the prediction performance would be affected if both positive and negative DMI values were included simultaneously—we note that, because our framework operates solely on the domain image (s_z) without any additional data, configurations generated with $+\tilde{K}_{DM}$ and $-\tilde{K}_{DM}$ would appear as nearly identical s_z textures carrying opposite (and unobserved) in-plane chirality. Including both signs in the training set would thus introduce a two-fold degeneracy in the input–output mapping: two distinct labels ($+\tilde{K}_{DM}$ and $-\tilde{K}_{DM}$) associated with essentially the same input image. Under a standard regression objective, this ambiguity drives the model toward the average of the two admissible solutions ($\hat{\tilde{K}}_{DM} \approx 0$), degrading rather than improving DMI identifiability. For the present study—intentionally constrained to the experimentally faithful s_z observable and starting only from the domain image without additional data—this sign degeneracy is therefore an intrinsic limitation of the input modality rather than of the estimation method, and we must presently accommodate it by restricting the sampling to $\tilde{K}_{DM} \geq 0$; the chirality handedness is instead fixed by convention through the DMI direction vector $\hat{\mathbf{d}}_{ij}$. This restriction, and the accompanying non-negative sampling range, are now stated explicitly in the Experimental Set-Up (Table 1 and the accompanying paragraph). Recovering the DMI sign would require chirality-sensitive observables (the in-plane components or the full three-dimensional spin field), which we identify as a direction for future work in the Conclusions.

These points are now supported in the revised manuscript by recent references: the out-of-plane sensitivity of the standard imaging modalities is backed by Li *et al.* (Interdisciplinary Materials, 2023), and the convention by which the sign of the antisymmetric exchange sets the chiral handedness by Fillion *et al.* (Nature Communications, 2022).

Point 2. *The manuscript appears to employ neural network architectures initialized* *from ImageNet-pretrained models. However, the features of natural images in ImageNet and magnetic domain images may have considerably different characteristics. For example, magnetic textures are mainly characterized by physical quantities such as correlation length, domain walls, and topological structures rather than object-like features. Could the authors clarify the motivation for using ImageNet-pretrained models in this problem? In addition, it would be helpful if the authors could discuss the difference between training randomly initialized networks and fine-tuning ImageNet-pretrained models, and clarify the role and advantage of pretraining in the present task.*

Response. We thank the reviewer for this pertinent question, which prompted us to re-examine the role of pretraining in our specific task. Our original motivation for initializing the backbones from ImageNet-pretrained weights was the standard transfer-learning expectation that generic low-level spatial filters (edges, gradients, and local textures) would accelerate and stabilize convergence during the early training epochs. However, we fully agree with the reviewer that the statistics of natural images and of magnetic domain textures differ substantially: magnetic configurations are characterized by physical quantities such as the correlation length, domain-wall morphology, and topological structure, rather than by the object-like semantic features that dominate ImageNet. This expectation should therefore not be taken for granted.

To address this point directly, we performed a controlled comparison in which all architectures were trained from random initialization (i.e., from scratch), and we additionally included a conventional (vanilla) CNN as a reference baseline. The results, now reported in the revised manuscript, show that ImageNet pretraining does *not* produce a meaningful improvement in the estimation of the Hamiltonian parameters: the marginal convergence advantage observed early in training does not translate into better final accuracy. Strikingly, the conventional CNN baseline—trained from scratch and devoid of any pretraining or specialized inductive biases—matches, and for some parameters even exceeds, the deeper residual, dense, depthwise, and attention-based backbones. This confirms that the estimation accuracy is governed by the physical information available in the s_z projection rather than by transfer learning from natural images.

In light of this finding, the updated workflow no longer relies on ImageNet-pretrained weights; all networks are initialized randomly and trained end-to-end on the magnetic dataset. This both simplifies the pipeline and removes the conceptual mismatch—rightly noted by the reviewer—between natural-image priors and magnetic-texture statistics. The revised manuscript reflects this change throughout: the *Pretraining* column of the architecture-configuration table (Table 2) now reports "None (from scratch)", the conventional CNN has been added to the architecture comparison (Deep Regression Framework, Section 3.3), and the Experimental Set-Up no longer invokes ImageNet pretraining.

Point 3. *In lines 406–407, the authors state that "This controlled thermal relaxation forces the continuous spin vector field to settle into the absolute global minimum." However, simulated annealing does not generally guarantee convergence to the absolute global minimum within a finite simulation time. This point is particularly important for DMI-driven magnetic systems, where many metastable states can exist due to chiral and topological energy barriers. Therefore, the expression "absolute global minimum" may be misleading. I suggest that the authors use a more careful description (or provide additional validation to support this statement).*

Response. We thank the reviewer for this important and entirely valid clarification. We fully agree that the expression "absolute global minimum" is misleading in the context of finite-time simulated annealing and have revised the manuscript accordingly.

As the reviewer correctly observes, simulated annealing provides a probabilistic guarantee of convergence to the global energy minimum only under a logarithmic cooling schedule ($T(t) \propto 1/\ln(t)$), which is computationally infeasible in practice. Our implementation employs a linear temperature decrement ($\Delta T = -0.1$ K) with 10,000 thermalization Monte Carlo sweeps followed by 2,000 measurement sweeps per temperature step, which constitutes a practical annealing protocol that promotes thorough exploration of the energy landscape but does not formally satisfy the conditions for guaranteed global convergence. This distinction is particularly relevant in DMI-driven magnetic nanodot systems, where the competition between symmetric exchange, antisymmetric exchange, and magnetocrystalline anisotropy generates a highly multimodal energy landscape populated by numerous metastable chiral and topological states separated by non-trivial energy barriers.

In the revised manuscript, we have systematically replaced all instances of "absolute global minimum" and "global energy minimum" with physically accurate terminology. Specifically, the outcome of the annealing process is now described as convergence to a *low-energy, thermodynamically stable configuration consistent with* *the annealing schedule*, rather than a guaranteed global minimum. This revision appears in:

- Section 3.2 (Monte Carlo simulation description, introductory paragraph);
- Section 3.2 (annealing convergence discussion, formerly Lines 406–407);
- Section 3.4 (RAMs description); and
- Section 5.1 (convergence analysis of the representative nanodot).

We believe this revised terminology accurately reflects the thermodynamic behavior of the simulation while avoiding any misleading implication of guaranteed global optimality.

Point 4. *Regarding Fig. 10, although I agree that it is not necessary to show every individual spin configuration in full detail, the current magnetic texture images appear too small to clearly identify the relevant features discussed in the manuscript. Improving the visualization would make the physical interpretation more accessible to readers.*

Response. We thank the reviewer for this helpful suggestion. We agree that, in the original version, the individual magnetization textures in the phase-diagram figure were too small to allow a clear identification of the relevant spin features. Following the reviewer's observation—and taking advantage of the reviewer's own remark that it is not necessary to display every individual spin configuration in full detail—we have replaced it with an improved version (now Figure 4 in the revised manuscript). The revised figure employs a coarser sampling grid with substantially enlarged representative magnetization images along the $T^{(0)}$–$\tilde{K}_{DM}$ axes, so that the characteristic textures—skyrmionic, labyrinthine/chiral, ferromagnetic, and paramagnetic—are now clearly discernible. This makes the physical interpretation of the phase transitions across the parameter space considerably more accessible to the reader, while preserving the overview of the structural diversity of the dataset.

REVIEWER 2

Point 1. *The novelty relative to prior work is not clear enough. The manuscript cites relevant studies, but it does not explain convincingly how this framework goes beyond earlier deep-learning approaches for Hamiltonian parameter estimation and magnetic-image inversion.*

Response. We thank the reviewer for this important remark, which led us to substantially strengthen both the related-work discussion and the explicit statement of our contribution. In the revised manuscript, the novelty is articulated along three concrete axes that jointly distinguish our framework from prior deep-learning approaches to Hamiltonian parameter estimation and magnetic-image inversion:

1. *Physical scale.* Whereas most existing approaches operate at the continuum micromagnetic scale, our framework is grounded in an *atomistic* description at the nanometric scale, based on a rigorous extended Heisenberg Hamiltonian that explicitly resolves the discrete spin interactions and the surface and finite-size effects governing magnetic nanodots.

2. *Simulation methodology.* Rather than relying on conventional classical simulation software, we derive a matrix/tensor (algebraic) decomposition of the extended Hamiltonian that recasts the atomistic energy as batched linear-algebra operations. Coupled with GPU computing, this formulation enables a massively parallel Metropolis–Hastings Monte Carlo with simulated annealing that accelerates dataset generation by orders of magnitude relative to sequential atomistic codes, without introducing any approximation to the underlying physics. The parallelization scheme (checkerboard sublattice updates, batched replicas, and multi-GPU execution) is now detailed in the Monte Carlo simulation subsection.

3. *Interpretability.* In contrast to the predominantly black-box models used in this field—which expose no mechanism to trace how spatial features drive the inferred parameters—we introduce Regression Activation Maps (RAMs), a novel error-based attribution method that supports layer-wise and structure-wise analysis of the regions relevant to each Hamiltonian parameter. RAMs are benchmarked against three perturbation-based methods—LIME, SHAP, and an exact spatial-frequency Shapley estimator (SF-SHAP)—through a deletion protocol, achieving competitive spatial faithfulness at a single backward pass

per explanation, as opposed to the hundreds to thousands of model evaluations required by these baselines.

To make these distinctions explicit, we have expanded the Related Work section, and we summarize the comparison in Table 1 below, which contrasts our framework against representative prior studies across six technical dimensions—the Hamiltonian terms modeled, the simulation method, the deep-learning architecture, the parameters estimated, the performance metrics, and, crucially, the interpretability mechanism—highlighting that these earlier works, while accurate, provide no spatially resolved physical explainability.

Table 1: Comparative analysis of deep learning approaches for magnetic Hamiltonian parameter estimation. The works of Kong *et al.* [6], Kwon *et al.* [7], and Lee *et al.* [8] are contrasted against the proposed framework across six technical dimensions.

Dimension	**Kong [6]**	**Kwon [7]**	**Lee [8]**	**Ours**
Hamiltonian terms	Exchange A, aniso. K_{eff}, DMI D	Exchange J, DMI β, dipolar D, aniso. K	Intralayer J, single-ion A, interlayer $J^{\perp}$	Ext. Heisenberg $\tilde{J}_1$–$\tilde{J}_4$, $\tilde{K}_{DM}$, $\tilde{H}_{ex}$, $\tilde{K}_{an1}$, $\tilde{K}_{anS}$
Simulation	Micromagnetic (mumax3, LLG)	MC simulated annealing	Atomistic (stochastic LLG)	Atomistic MC (Metropolis + annealing, multi-GPU)
DL architecture	Custom CNN	ResNet18/50, CNN	MLPs	CNN baseline + DenseNet, ViT, Xception, ResNet
Params. estimated	4	3	3	8
Metrics	MAE, R^2	MAE, PCA of errors	MAPE, MSE	R^2, MAE + UMAP/HDBSCAN clustering
Interpret.	Feature maps (black box)	Black box	Black box (PCA)	**RAMs** (layer/structure) + identifiability

Point 2. *The claim that eight Hamiltonian parameters can be recovered is over-*

stated. According to the main results, J3, J4, the anisotropy parameters, and the Zeeman field show low or near-zero identifiability from the two-dimensional Sz projection.

Response. We thank the reviewer for this valid and important observation, which prompted a substantial revision of both the dataset and the results. In the original submission, the anisotropy constants and the Zeeman field indeed showed low identifiability, because the external field was applied exclusively in-plane and the anisotropy was sampled over a narrow range—so that these parameters coupled predominantly to the unobserved in-plane spin components. We agree that, under those conditions, claiming the recovery of eight parameters was overstated.

To address this rigorously, we regenerated the dataset with two deliberate physical enrichments: (i) the external field is now applied along the out-of-plane axis in part of the simulations, which couples the Zeeman term directly to the observable s_z component, and (ii) the magnetocrystalline anisotropy range is extended toward high values, imprinting sufficiently distinct domain-wall morphologies on the out-of-plane texture. The new dataset comprises 166,141 configurations, and all models were retrained and re-evaluated on it.

The updated results, now reported in the revised manuscript, show that six of the eight parameters are recovered with high fidelity from the s_z projection: the DMI strength ($R^2 \approx 0.99$), the annealing temperature ($R^2 \approx 0.96$), the external Zeeman field ($R^2 \approx 0.95$), the bulk anisotropy ($R^2 \approx 0.94$), and the second-shell exchange ($R^2 \approx 0.89$), with the surface anisotropy at an intermediate level ($R^2 \approx 0.65$). Crucially, we no longer claim the recovery of all eight parameters: the two higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ retain near-zero R^2 across all architectures and all magnetic phases. We now present this explicitly as a *fundamental identifiability limit* of the inverse problem—these constants are energetically dominated by the first- and second-shell exchange and therefore leave no distinguishable signature in the s_z texture—rather than as a recoverable quantity. This honest characterization of what is and is not identifiable from the two-dimensional projection is described in the Experimental Set-Up (new dataset and sampling ranges), quantified in the Results (per-parameter and per-phase R^2/MAE), and reflected in the identifiability discussion.

Point 3. *The generalization test is weak. The training, validation, and test sets are randomly split from the same synthetic dataset, all generated by the same Monte*

Carlo pipeline. This mainly tests interpolation within one simulator, rather than true generalization to new physical regimes, geometries, imaging conditions, or materials.

Response. We thank the reviewer for this thoughtful comment. The training, validation, and testing partitions follow the hold-out protocol commonly adopted for image-based regression, with a fixed random seed to guarantee reproducibility, which is the same evaluation standard used by the reference deep-learning studies for magnetic Hamiltonian parameter estimation that we cite. We would, however, respectfully clarify that our evaluation is not merely an interpolation test within a single homogeneous distribution. The dataset deliberately spans a broad range of physically distinct regimes—from ordered helical, skyrmionic, and Ising-like textures to thermally disordered and mixed-chirality (bimeron) configurations, and across a wide temperature range that includes transient and paramagnetic states. To make this explicit, we have added a per-phase analysis in which the unsupervised clustering of the latent space isolates each magnetic phase, and we report the regression accuracy separately for every phase. This demonstrates that the model generalizes across qualitatively different physical regimes rather than exploiting artifacts of a single texture family, and it transparently exposes which regimes remain challenging (the disordered and mixed-chirality phases).

We fully acknowledge, however, that a direct quantitative validation on experimentally acquired magnetization images lies beyond the scope of the present study, which is deliberately framed around simulated data. Bridging this simulation-to-experiment gap—through domain adaptation, physics-informed data augmentation, or few-shot transfer learning from experimentally labeled samples—is explicitly identified as a direction for future work in the revised Limitations and Conclusions, and we consider it the natural next step to establish generalization beyond the simulator.

Point 4. *The parameter sampling strategy needs stronger physical justification. The eight parameters are sampled independently and uniformly, but real materials do not necessarily span such an independent parameter space. The authors should explain whether these sampled combinations are physically realistic, and whether the model is learning transferable physical relationships rather than artificial correlations in the synthetic dataset.*

Response. We thank the reviewer for this important observation regarding the physical realism of the sampling strategy and the risk of learning artificial correlations. We address both concerns explicitly.

Each Hamiltonian parameter is drawn within physically motivated bounds using a quasi-random (low-discrepancy) sampling scheme rather than naïve uniform sampling, which ensures a systematic and homogeneous coverage of the admissible parameter region. We computed the full pairwise correlation matrix of the eight parameters over the entire dataset of 166,141 configurations, shown in Figure 1 below: the vast majority of parameter pairs are essentially uncorrelated ($|\rho| \lesssim 0.3$), with a single appreciable correlation between the second-shell exchange $\tilde{J}_2$ and the DMI strength $\tilde{K}_{DM}$ ($\rho \approx -0.53$).

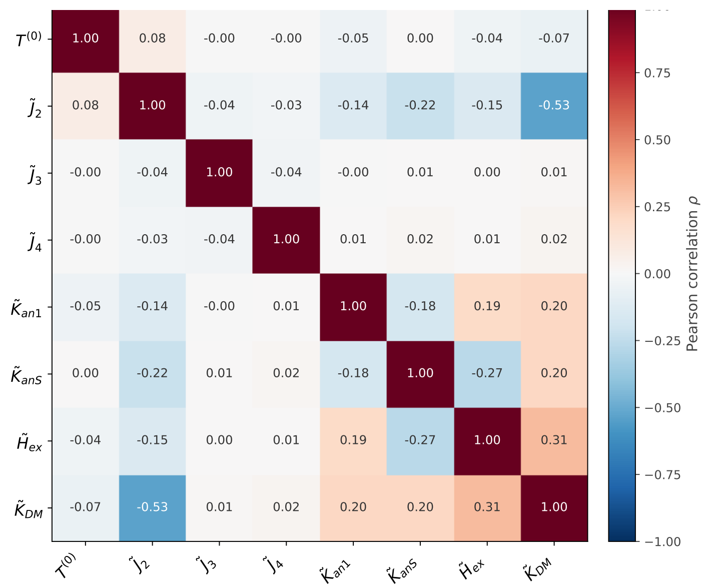

Figure 1: Pairwise Pearson correlation matrix of the eight Hamiltonian parameters computed over the entire dataset of 166,141 configurations. All parameter pairs are essentially uncorrelated ($|\rho| \lesssim 0.3$), with the single exception of $\tilde{J}_2$ and $\tilde{K}_{DM}$ ($\rho \approx -0.53$), whose anti-correlation is a direct consequence of the deliberate design in which the exchange-frustration and chiral mechanisms are activated in a mutually exclusive fashion. The absence of spurious cross-parameter correlations indicates that the model cannot exploit artificial statistical shortcuts to infer one parameter from another.

Crucially, this correlation is not an artifact of biased sampling but the direct signature of a deliberate physical design. Within our single unified dataset, we span two complementary mechanisms capable of stabilizing modulated (non-collinear) magnetic textures: an *exchange-frustration* regime, in which the second-shell exchange

$\tilde{J}_2$ is varied while the DMI is switched off, and a *chiral* regime, in which the DMI $\tilde{K}_{DM}$ is varied while $\tilde{J}_2$ is switched off. The scientific motivation was precisely to test the hypothesis that exchange frustration can drive spin textures analogous to those induced by the antisymmetric DMI—i.e., that these two physically distinct interactions constitute alternative routes to similar modulated states. Activating the two mechanisms in a mutually exclusive fashion is what allows their contributions to be cleanly isolated, and this design choice—rather than any hidden statistical artifact—is what produces the $\tilde{J}_2$–$\tilde{K}_{DM}$ anti-correlation. Within each regime, the active parameter spans its full physical range, so that the network must infer the *magnitude* of the interaction from the magnetization texture rather than merely its presence or absence.

Regarding whether the model learns transferable physical relationships rather than dataset-specific correlations, we offer two lines of evidence. First, the network recovers both $\tilde{J}_2$ ($R^2 \approx 0.89$) and $\tilde{K}_{DM}$ ($R^2 \approx 0.99$) with high fidelity, which means it successfully discriminates between two distinct physical origins of visually similar modulated textures—the opposite of exploiting a trivial correlation. Second, and most directly, the Regression Activation Maps (RAMs) introduced in this work show that the model bases its predictions on physically meaningful spatial features (skyrmion cores, chiral domain walls, stripe periodicity) rather than on dataset-level statistical shortcuts, providing explicit, spatially resolved evidence that the learned representations are physically grounded.

Regarding the physical realism of the sampled ranges, we have added a dedicated justification to the revised Experimental Set-Up. Because $\tilde{J}_1$ is fixed as the energy reference, the physically meaningful quantities are the dimensionless ratios ($\tilde{J}_r/\tilde{J}_1$, $\tilde{K}_{DM}/\tilde{J}_1$, $\tilde{K}/\tilde{J}_1$), which we show fall within the ranges reported by first-principles calculations and experiments for chiral magnets and ultrathin transition-metal films: the decaying, sign-alternating higher-order exchange reproduces density-functional results for Fe/Ir(111) and Pd/Fe/Ir(111) [1, 2]; the anisotropy range spans from soft magnets to strongly anisotropic interfacial systems [4, 5]; and the Zeeman range maps, via $E = g\mu_B B$, onto laboratory fields up to ≈ 10 T. We also address the one apparent exception: our DMI ratio $\tilde{K}_{DM}/\tilde{J}_1$ reaches ≈ 1.2, higher than the ~ 0.3–0.7 of typical interfacial materials. This is deliberate and standard in finite-lattice Monte Carlo studies, where a larger ratio ensures that the helical period fits several times within the simulation cell [3]; because the physics is scale-invariant, a large ratio on a small lattice is equivalent to a small ratio in a large system, so this reflects finite-size rescaling rather than unphysical values.

Finally, we agree that a natural next step is a dataset in which $\tilde{J}_2$ and $\tilde{K}_{DM}$ co-vary simultaneously, as may occur in real materials where both interactions coexist; we now identify this as a direction for future work, in line with the generalization discussion of the previous point.

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